

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)**MID SEMESTER EXAMINATION****SUMMER SEMESTER, 2023-2024****DURATION: 1 HOUR 30 MINUTES****FULL MARKS: 75****CSE 4849: Human Computer Interaction**

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Suppose you are developing a mental health therapy chatbot designed to provide conversational support to users experiencing stress, anxiety, or depression. The chatbot engages in dialogue by detecting emotional cues from user messages, maintaining context across multiple sessions, and referring back to past interactions. It draws on a knowledge base of therapeutic techniques, including cognitive behavioral therapy, mindfulness, and active listening strategies. It evaluates the users progress, adjusts its responses, and recommends coping strategies based on patterns it has observed and user feedback. Answer the following:
 - i. Map the chatbot's actions to the cognitive processes of perception, attention, memory, and knowledge. Explain how these processes are simulated to enable meaningful and empathetic mental health conversations. 7
(CO1)
(PO1)
 - ii. The chatbot is designed to simulate reflective cognition, providing personalized and adaptive support. How can reflective thinking be built into the system? 8
(CO1)
(PO1)
- b) Consider the Cognitive Architecture for Language Agents (CoALA) in Figure 1 proposed by Sumers et. al. from Princeton University. Identify and design each CoALA module in the system by creating an interaction flowchart showing how the chatbot would process, remember, adapt, and act during a single therapy session. 10
(CO2)
(PO3)

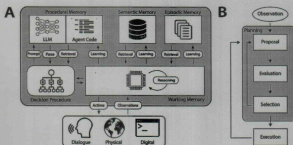


Figure 1: Cognitive architectures for language agents (CoALA). A: CoALA defines a set of interacting modules and processes. B: Temporally, the agents decision procedure executes a decision cycle in a loop with the external environment.

2. a) In human visual perception, cone cells handle central vision, which is responsible for recognizing fine details, while rod cells manage peripheral vision, detecting motion and broader visual cues. Our ability to distinguish fine detail decreases as objects move away from our central vision. This principle has design implications that help ensure the app is both effective and safe. Consider the following list of tasks while designing a map navigation app for the car drivers: 10
(CO2)
(PO2)

- Reading Estimated Time of Arrival (ETA)
- Noticing a change in the color of the route line (e.g., red for heavy traffic)
- Following Turn-by-Turn text instructions
- Observing incoming call or message notification banners during navigation
- Tapping the screen to search for a new destination

For each task, consider the design implications and decide whether it should be designed for central vision (cone cells) or peripheral vision (rod cells). Provide a one-line justification for your choice.

- b) In a graphical user interface, a pointing task involves four targets labeled as A, B, C, and D, placed at varying distances from the starting point. The distances (D) from the starting point to the center of each target are as follows: $D_A = 180$ pixels, $D_B = 160$ pixels, $D_C = 180$ pixels, and $D_D = 140$ pixels. The widths (W) of the target areas are: $W_A = 20$ pixels, $W_B = 18$ pixels, $W_C = 80$ pixels, and $W_D = 15$ pixels. Constants for movement time (MT) calculations are, $a = 90$ ms, $b = 40$ ms. Calculate human throughput using Fitts' law for each target and determine which target provides the highest throughput. 15 (CO2) (PO1)
3. a) Bangla word-level sign language, 'BdSLW60' dataset, contains 60 isolated Bangla sign language words with 9307 annotated gesture trials from 18 signers. Each sign is captured as a video and includes variations in signer style, hand shape, and motion. 13 (CO2) (PO2)
- Gestures are described using landmark coordinates (hand, face, body) and temporal parameters like speed, duration, and direction.
- To represent the meaning and structure of each gestural word, we can use techniques such as semantic networks, frame-based, script-based, or rule-based knowledge representation. For example, the sign for "Abba (Father)" involves raising the right hand near the forehead, moving it forward slightly with fingers open. The gesture lasts about 1.2 seconds and starts from the right shoulder, with the palm facing inward.
- Which knowledge representation technique (e.g., semantic network, frame-based, script-based, or rule-based) is most appropriate for modeling the signs? Justify your answer with reasoning and briefly describe how the gesture would be represented. [Hint: You can use a JSON-like structure to represent the gesture and its key parameters]
- b) You are hired as an interaction designer to evaluate user interactions in a newly developed fitness tracking app. This app allows users to record workouts, monitor daily steps, and manage personal goals using touch gestures on smartphones. Users input commands through the app, and the system provides feedback on their activities. Based on the Abowd and Beale interaction framework, identify and explain the interaction problems (articulation, performance, presentation, and observation) that may exist (one or more) for the following tasks within 1-2 line justification for each problem you found. 12 (CO3) (PO1)
- Users try to log their workout sessions by selecting exercise types and durations, but some users find it difficult to select precise exercise categories, leading to inaccurate workout logs. There are also occasional delays when saving session data.
 - Users navigate to view their weekly step count and goal progress. Some users report that the graphs and statistics are confusing and hard to interpret, making it difficult to understand if they are meeting their fitness goals.
 - Users attempt to set reminders for daily hydration but miss seeing the confirmation alert after enabling reminders, causing uncertainty about whether the reminders were successfully set.
 - Users report that when starting a live workout tracking session, the app occasionally freezes or crashes, forcing them to restart the app and lose their current session data.